Bias & Varience

2400032783
P. Sathvik

→ Bias is Error from wrong assumptions in the model.

→ High bias → model is too simple.

→ Misses important patterns.

→ leads to Underfitting.

Example :- Using a straight line to model a clearly curved relationship.

Symptoms :-

→ High training error.

→ High test error.

→ Model barely learns.

→ Varience :-

Varience is error from too much sensitivity to the training data.

→ High Varience → model is too complex.

→ learns noise instead of signal.

→ leads to overfitting.

Ex :- A deep decision tree that memorizes every data point.

2400032783
P. Sathvik

Symptoms :-

→ Very low training error

→ High test error

→ Model changes a lot with small data changes.

Model Type	Bias	Varience	Result
Too Simple	High	low	Underfitting.
Too Complex	low	High	Overfitting.
Balanced	low	low	Best generalization

Generalization :-

Generalization = how well a model peforms on unseen data.

A model generalizes well when :

→ Training error ≈ Test error.

→ Both errors are resonably low.

→ High bias → poor generalization.

→ High varience → poor generalization.

→ Balanced Bias & varience → good generalization

P. Sathvik
2400032783

Case study :- House price prediction using machine learning.

Problem :-

A real estate Company wants to predict house prices based on :-

→ size

→ No of bedrooms.

→ location score.

→ Age of houses.

→ The experiment with different models and observe bias, variance, generalization effects.